

FOLDER LOCK Two-Factor Folder Encryption Face Recognition → Password → Unlock

Executive Summary

FOLDER LOCK converts an ordinary folder on a professional's computer into an encrypted vault. Access to the vault requires successful authentication using two independently validated factors. Layered on top of the encryption engine is an offline AI assistance component. The AI runs entirely on the local machine.

The result is a self-contained, single-click security tool that a Chartered Accountant can deploy on an office or laptop machine in under two minutes, that survives laptop theft, and that produces a tamper-evident audit trail acceptable for internal quality review and provides security controls that may support appropriate technical and organisational measures adopted by the organisation for protection of personal data under the applicable data-protection framework.

Problem Statement

A working professional's laptop typically carries, in plain unencrypted form, a set of folders such as:

- Client-wise income tax folders containing PAN cards, Aadhaar copies, Form 16, Form 26AS, capital gains statements and bank statements.
- GST working folders with purchase and sales registers, e-way bill data and reconciliation workings.
- Audit folders with trial balances, ledgers, management representation letters, draft financial statements and working papers.
- Bank and stock audit folders with borrower data, sanction letters and inspection notes.
- Payroll and TDS folders with employee salary details, PAN and bank account numbers.

These folders are ordinarily protected by nothing more than the Windows login password. Anyone who obtains the machine, boots it from external media, removes the hard disk, or simply sits at an unlocked desk during lunch hour has complete access to every client file.

Why existing options do not solve the problem - Existing tools either protect the whole disk but not individual folders, or protect one file but not the folder around it and none of them ask for a second key.

Existing option	What it does	Where it fails
Windows login password	Prevents interactive login to the user profile.	Files remain unencrypted on disk. Removing the disk, or booting a live USB, exposes everything. Offers zero protection against theft.
ZIP / RAR password protection	Encrypts an archive with a password.	Single factor. A leaked or guessed password is total compromise. No audit trail, no automatic re-lock, and the user must re-zip the whole archive after every edit.
Office file passwords	Password-protects individual Excel or Word files.	Applies to one file at a time, is routinely shared over WhatsApp and e-mail, and does not cover PDFs, scans, JSON or Tally exports sitting in the same folder.
BitLocker / full-disk encryption	Encrypts the entire drive at rest.	Once Windows boots, every folder is fully readable to anyone at that logged-in session. It does not separate one client's data from another, is unavailable on Windows Home editions, and gives no per-folder audit trail.
Cloud drives with 2FA	Two-factor protection at the cloud account level.	The synced local copy on the laptop is plain text. Also introduces confidentiality and data-residency concerns for client records.

Objective

Primary objective is to make the selected folder cryptographically inaccessible while the vault is locked, and require successful two-factor authentication before the application exposes the protected contents to the authenticated user.

Technology Used

1. Comes with Python — nothing to install

Component	What it is used for
Python 3.12 or later	The language the whole application is written in.
tkinter / ttk	The screens you see — main window, menus, toolbar, vault list, dialog boxes, progress bar, status bar.
sqlite3	A small local database holding vault settings and the record of access.
pathlib	All handling of files and folders, including names with Indian-language characters.
logging	Diagnostic logs that roll over automatically so they never fill the disk. They never contain file contents or keys.
configparser	The settings file (config.ini) — idle timeout, default vault location, alert thresholds, assistant on or off.
threading / queue	Keeps the screen responsive while a large folder is being locked, and runs the idle re-lock timer.
secrets / os.urandom	Generating keys, salts and the recovery code in a genuinely unpredictable way.
hashlib / hmac	File integrity fingerprints and the chaining of the access record.

2. Tools used

Tool	Used for
ChatGPT	Drafting and refining the development instructions.
Claude (Opus)	Architecture, writing and reviewing Python code, threat modelling, tests and documentation.
System requirement	Windows 10 or Windows 11

Output

- The complete application, with all source code.

- A settings file (config.ini) with sensible defaults that you can edit in Notepad.
- A README written for a user, not for a programmer: what it does, how to install it, how to use it.

Process to use -

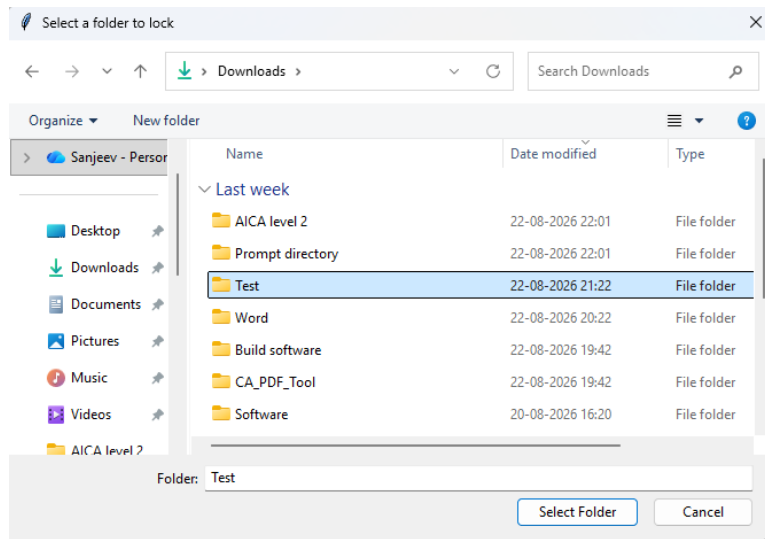
- Turns any chosen folder into a locked safe. The folder stays in the same place with the same name — but its contents become unreadable characters.
- Opens only after both keys are given. Takes about ten seconds.
- Re-locks by itself after fifteen minutes of inactivity, and when the computer shuts down.
- Keeps one vault per client, so opening one does not expose the others.
- Runs entirely on your own computer. No cloud, no subscription, no internet after the first run

Snapshot of program run:

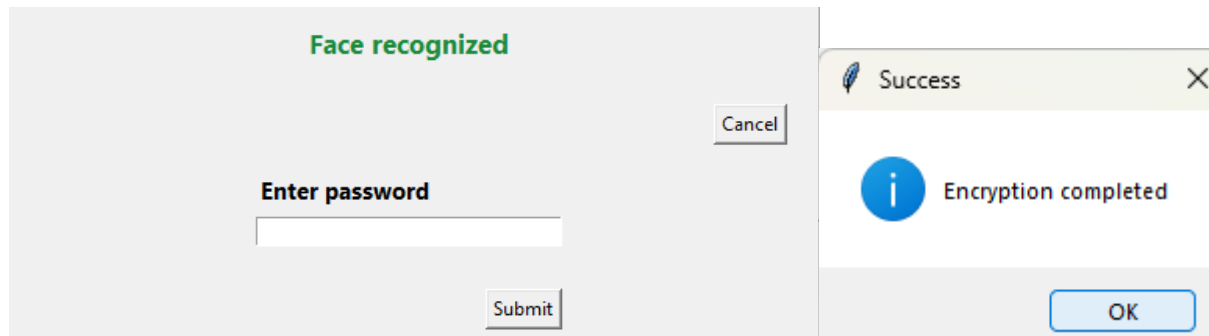
- GUI



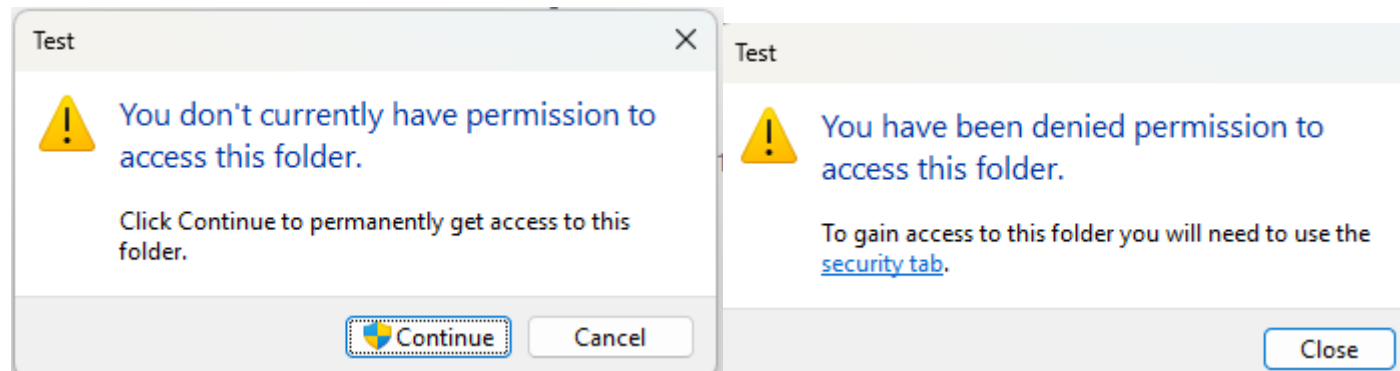
- Selection of folder



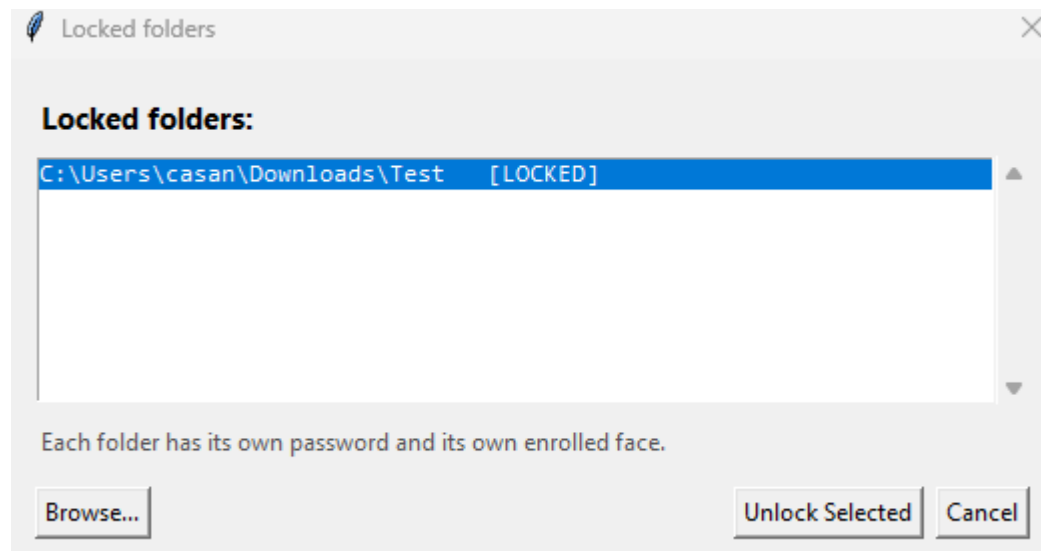
- Password recognition



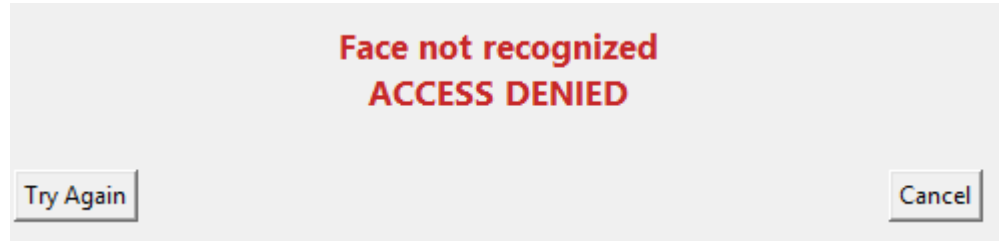
- Test whether folder can open without password



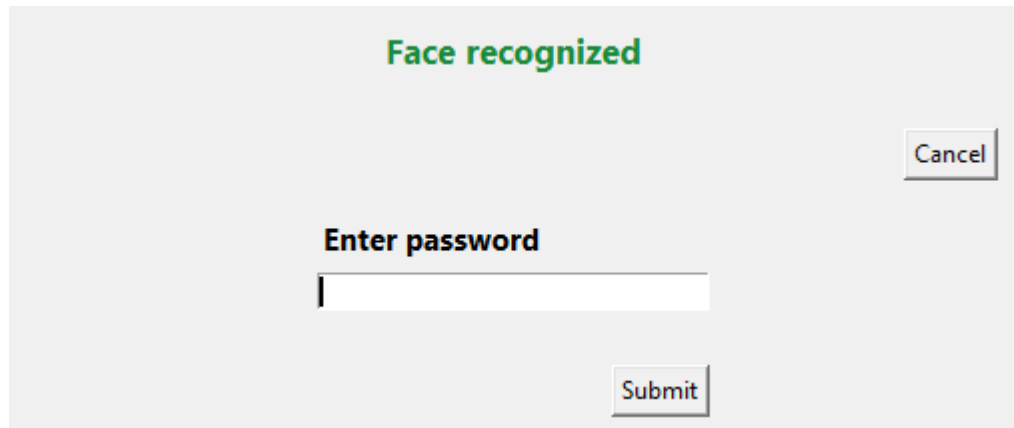
- Unlocking of folder



- Trial to verify whether face authentication incorrectly grants access to an unregistered face



- Face authentication



- Unlocked

